

CS309 Lab 5 ARM Advance Program Gasoline Pump

Objective: The objective of this assignment is to have the students build upon the skills from ARM Lab 4 for a more complex ARM Assembly program. The students will have to track multiple items forcing the use of memory or the stack to save and restore data values.

Your program will simulate operation of a gasoline pump.

The pump will dispense three (3) types of gasoline, regular, mid-grade, premium. The price per gallon (in cents) is given in the following table.

Grade	Price per gallon in cents	For every dollar dispense this amount of gasoline in tenths of gallons.
Regular	250	4
Mid-Grade	333	3
Premium	500	2

The initial inventory level is 500 tenths of gallons (50 gallons) per grade. You should keep all values (gallons, price per gallon and dispense amount) in integer format. Note: based on the numbers above it is not possible to dispense all the Mid-Grade gasoline.

When the program starts it shall display an identification and the current inventory level. For example:

Welcome to gasoline pump.
Current inventory of gasoline (in tenths of gallons) is:

Regular 500
Mid-Grade 500
Premium 500

Dollar amount dispensed by grade:

Regular \$0
Mid-Grade \$0
Premium \$0

The program shall then prompt the user for the grade to dispense and the selection displayed. For example:

Select Grade of gas to dispense (R, M or P)
R
You selected Regular.

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The pump shall then prompt the user for the dollar amount to dispense. The program shall reject any invalid dollar amounts (at least \$1 and no more than \$50) and any non-integer inputs. For example:

Enter Dollar amount to dispense (At least 1 and no more than 50)
10
40 tenths of gallons have been dispensed.

Prior to dispensing gasoline, the pump shall ensure there is enough gasoline to meet the request. If there is not enough gasoline to meet the request the user shall be notified and prompted to enter a new (lower) dollar amount.

If any grade of gasoline drops below 1 gallon (10 tenths of gallons) consider it to be an insufficient amount to dispense and do not allow the user to select that grade to dispense. When all three grades drop below 1 gallon each then display final inventory levels and dollar amounts dispensed by grade and shut the pump off (i.e., stop the program and return control to the operating system).

Your program should check for and reject any invalid inputs for grade selection and dollar amounts.

The pump shall allow the entry of a hidden code at the main prompt (i.e., Select Grade). When this hidden code is entered the pump shall display the inventory level of gasoline and the dollar amount dispensed by grade so far.

Pework:

Students need to have a file which, at the very least contain the following:

1. Proper header information that include commands that assemble, link, run the file and run the file using the debugger that is specific for their file.
2. Data section that defines all the output strings.
3. Code comment sections that describe the major sections of the code with the output loop clearly identified.

If the student already has a working program this prework check is not required.

This program was assigned:

Spring 2021
Spring 2020
Summer 2019
Summer 2017
Spring 2022

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Student _____

Lab Instructor: _____

Total Score: _____

Late Penalty? No _____

Yes: _____

Feature	Points off
Code Comments/Documentation - Class, Term, Author, Date. - Purpose of software - Documentation for the start of each loop. - Documentation for error checking - In-line comments that explain why the following code exists.	
Welcome message and instructions displayed and are clear.	
The initial inventory of gas and money is displayed correctly.	
The program rejects invalid inputs for the gasoline grade selection. Any input other than R, M, P or secret code.	
Select R then enter the following invalid entries for dollar amount: • -10 (integer < 1) • 55 (integer > 50) • X (character)	
With grade of R still selected enter 10 (valid amount). Verify 40 tenths of gallons is dispensed.	
Select grade of M then enter 20. Verify 60 tenths of gallons is dispensed.	
Select grade of P then enter 30. Verify 60 tenths of gallons is dispensed.	
Enter secret code and verify the following inventories: - Reg 460 - Mid 440 - Prem 440 - Dollar Amounts: - Reg \$10 - Mid \$20 - Prem \$30	
For each grade of gas dispense \$50 for each grade and verify the following amounts are dispensed: - Reg – 200 - Mid – 150 - Prem - 100	
Repeat the above inputs once and ensure the correct amounts are dispensed.	

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Enter secret code and verify the following inventories: Reg 60 Mid 140 Prem 240 Dollar Amounts: Reg \$110 Mid \$120 Prem \$130	
Select Regular grade and enter \$20. The program should display insufficient inventory and prompt for new lower amount. Enter \$15. The program should dispense 60. Note there is no more Regular grade gas at this point.	
Select Mid-grade and enter \$50. The program should display insufficient inventory and prompt for new lower amount. Enter \$45. The program should dispense 135. Note there is insufficient Mid-Grade gas to dispense any more.	
Select Premium grade and enter \$50. The program should dispense 100.	
Select Premium grade and enter \$50. The program should dispense 100.	
Enter secret code and verify the following inventories: Reg 0 Mid 5 Prem 40 Dollar Amounts: Reg \$125 Mid \$165 Prem \$230	
Select Regular grade and ensure program states there is insufficient inventory and prompt for new grade.	
Select Mid-grade and ensure program states there is insufficient inventory and prompt for new grade.	
Select Premium grade and enter \$20. The program should dispense 40. At this point all grades are below 1 gallon. The program should display the following inventories: Reg 0 Mid 5 Prem 0 Dollar Amounts: Reg \$125 Mid \$165 Prem \$250 The program should then terminate and return control back to the OS.	